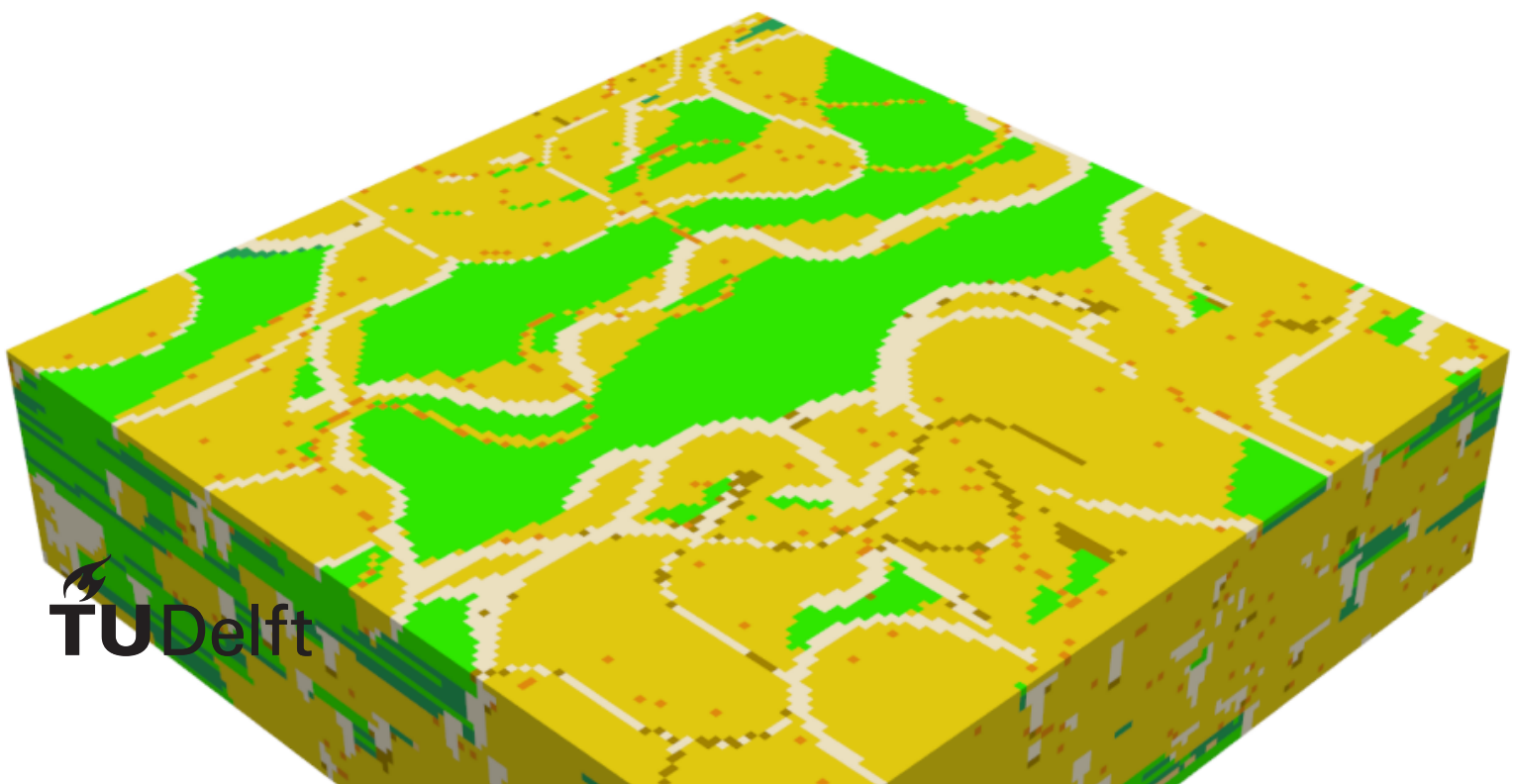


Evaluating the applicability of Generative Adversarial networks to generate geological facies models

A Case study for the Delft Area

M.D. Torres Ruhe



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Abstract

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Acknowledgements

I would like to thank...

List of Abbreviations

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cGAN Conditional Generative Adversarial Network. 11

DSST Delft Sandstone. 3, 4

FluvGan Fluvial Generative Adversarial Network. 11

GAN Generative Adversarial Network. 11

TUD Technische Universiteit Delft. 3

TvD True Vertical Depth. 3

WNB West Netherlands Basin. 3

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1 Introduction

Sedimentary deposits represent a vital commodity in modern society due to the various resources that can be extracted from them. These resources include minerals, groundwater, geothermal heat, sands and hydrocarbons (e.g., Brekke et al., 2001; Donselaar et al., 2017; Donselaar et al., 2015; Morris & Ramanaidou, 2007). Moreover, recent studies have investigated storage possibilities for carbon dioxide and hydrogen in this subsurface environment (Heinemann et al., 2021; Zhou et al., 2020). Beyond the increase in subsurface applications, the world also deals with an increasing population and energy demand. Consequently, the demand for resources hosted within fluvial sedimentary deposits is expected to increase while the distribution of the commodities themselves becomes sparser and of lower quality (e.g., Calvo et al., 2016; Spittler et al., 2020). This highlights the need for a deeper understanding of subsurface environments to enable more precise resource management which can be achieved through various methods such as: data gathering surveys, geological evolution and constructing high-quality 3D geomodels.

3D geomodels are traditionally constructed by engineers and geoscientists to predict reservoir properties and resource distribution (Royer et al., 2015; SONG et al., 2022). These predictions are fundamentally based on a foundational geological conceptual model. The conceptual model acts as a hypothesis regarding the most likely rock mass configuration, built entirely from sparse field data (Koltermann & Gorelick, 1996). This data sparsity is an unavoidable reality of subsurface exploration, resulting from the exceptionally high costs and extended runtimes required for physical data acquisition. Consequently, engineers must develop a working understanding of the subsurface based on highly limited information. This reliance on limited information drives the continuous need to improve geomodelling techniques. Such improvements aim to computationally bridge the gap between isolated data points and realistic, continuous geological representations.

1.1. Problem definition

Despite ongoing advancements in geomodelling techniques, the subsurface environment remains inherently characterized by a lack of data. This data scarcity dictates that functional 3D geomodels must rely heavily on subjective interpretations of the underlying geological conceptual model. Because transitioning from these interpretive hypotheses into a digital framework requires applying specific mathematical and spatial algorithms, this workflow inevitably introduces uncertainty into the final model. This inherent uncertainty originates at multiple stages, stemming from both the sparse data inputs and the modelling steps themselves (Høyer et al., 2024). Although this uncertainty is an inescapable aspect of subsurface engineering, precisely measuring its extent is crucial for effective decision-making. This quantification is usually accomplished via Monte Carlo simulations, necessitating the creation of a large collection of equally likely 3D geomodels. Creating this extensive ensemble, however, reveals a core drawback in conventional modeling techniques: the significant compromise between computational efficiency and geological precision. To sustain the generation rate needed for extensive uncertainty quantification, traditional methods frequently depend on greatly simplified, weak geological assumptions.

Due to the fact that these approaches are often limited to simple statistical distributions or strict object-oriented rules (Rongier & Peeters, 2025b), they find it challenging to replicate the intricate, non-linear architectural relationships present in natural sedimentary formations (Sun et al., 2023). As a result, existing workflows compel engineers to decide between models that can quickly quantify uncertainty but do not possess geological realism, or highly precise multi-conceptual models that require too much effort to produce in large quantities (Enemark et al., 2019). This bottleneck highlights a significant deficiency in contemporary subsurface characterization. A significant need exists for a modeling approach that can thoroughly incorporate complex geological prior knowledge while also ensuring the fast generation rates necessary for ensemble-based uncertainty quantification. As a result, the inquiry comes

up regarding how the existing modelling process can be modified to ensure that:

- 1) Previous geological insights and structural intricacies are better employed.
- 2) Uncertainty in the produced models is quantified more precisely and efficiently.

To address the need for both geological realism and rapid ensemble generation, recent research has proposed advanced machine learning algorithms as a highly effective solution. These algorithms, specifically the modern subset known as ([Deep Generative Modelling \(DGM\)](#)), possess the unique ability to learn complex, non-linear spatial relationships directly from data. By learning these intricate relationships, [DGM](#) can accurately capture the geological complexity and non-stationarity that traditional methods often miss ([Rongier & Peeters, 2025b](#)). Furthermore, once such a model is fully trained and validated, it can generate a vast ensemble of geologically realistic 3D realizations almost instantaneously. This rapid generation directly facilitates the large-scale uncertainty quantification demanded by modern subsurface engineering.

1.2. Research gap and opportunity

While the broader application of machine learning in geomodelling has historical roots ([Demyanov et al., 2008](#)), recent surges in computing power have yielded highly capable 3D [DGM](#) architectures, including [Variational Auto Encoders \(VAEs\)](#) and Latent Diffusion Models (e.g., [Bhavsar et al., 2024](#); [Di Federico & Durlofsky, 2025](#); [Laloy et al., 2017](#); [Song et al., 2023](#)). Among these varied architectures, this study specifically focuses on Generative Adversarial Networks ([Generative Adversarial Networks \(GANs\)](#)). While recent literature highlights Latent Diffusion Models as a vanguard for precise hard data conditioning, they inherently suffer from slow inference speeds due to their iterative denoising process. In contrast, [GANs](#) offer near-instantaneous sampling speeds perfectly suited for large-scale Monte Carlo uncertainty loops, despite well-known training vulnerabilities such as mode collapse. [GANs](#) are therefore selected due to their computational efficiency, established research history in spatial generation, and the wide availability of architectures strictly tailored for geomodelling. However, successfully training such [GANs](#) requires an extensive volume of high-quality data to act as the geological prior. To overcome the inherent scarcity of real subsurface data, process-based forward modelling software, such as FLUMY ([Martin et al., 2016](#)), can be utilized to generate robust synthetic training datasets.

Utilizing these synthetic datasets allows [GANs](#) to effectively learn geological priors; however, a critical research gap remains regarding their ultimate deployment. While [DGM](#) excels in theoretical environments, recent literature emphasizes an urgent need to transition from idealized, synthetic test cases to practical field applications. As [Rongier and Peeters \(2025b\)](#) stated, “We now need to move towards more detailed studies on larger case studies closer to field applications to further assess how the whole workflow performs”. Similarly, [Di Federico and Durlofsky \(2025\)](#) noted that the application (and extension as required) of the overall methodology to real cases should be performed. Addressing this explicit mandate forms the core opportunity of this research. Consequently, the primary goal of this project is to bridge the divide between synthetic training environments and practical field operations by assessing the performance of deep generative modelling in a more complex, real-world application.

Testing these models in a real-world application requires targeting an active subsurface system, a role fulfilled in this study by utilizing data from the Delft Sandstone reservoir. This reservoir target must be characterized using a comprehensive geological conceptual model combined with sparse well data. Such sparse well data will serve as the hard conditioning constraints to evaluate how the generative model behaves under tight physical limitations. These limitations, and the resulting fluctuations in uncertainty, will be analyzed by testing the workflow across multiple distinct geological priors. Investigating the interplay between varied priors and sparse conditioning constraints ultimately establishes the framework for the specific objectives of this study.

1.3. Objective and research questions

The main goal of this thesis is to assess the practical feasibility of using a **GAN** to generate models of a real-world reservoir by conditioning it to well data. The outcome of this thesis may indicate whether GANs can be used to model the subsurface in various settings and if other **DGM** techniques should be considered. Hence, the main research question of this work is: **To what extent can Deep Generative Modelling (DGM) be used to create geologically plausible realizations of the Delft Sandstone reservoir while conditioned to sparse real-world data?**

To help answer this question, several subquestions are formulated:

1. How should the Delft Sandstone's geology be parameterized in FLUMY to produce a training dataset sufficiently diverse for DGM modeling?
2. What type of **GAN** architecture best suits facies modelling?
3. How well is the model able to reproduce the complexity inherent in the training data?
4. How well does the optimized model produce geologically plausible structures?
5. How different do models trained on different geological priors react when sparse data is introduced?

1.4. Thesis outline

The remainder of this thesis is structured to systematically address these research objectives. Chapter 2 establishes the theoretical foundation by reviewing the principles of geological modelling, the mechanics of deep generative models, and the specific geological setting of the Delft Sandstone Member. Building upon this theoretical background, Chapter 3 outlines the core workflow, describing the generation of synthetic training datasets using FLUMY and detailing the specific Generative Adversarial Network (**GAN**) architecture deployed for the reservoir modelling tasks. The empirical outcomes of this workflow are presented in Chapter 4, which evaluates the training process and analyzes the structural and spatial realism of the generated realizations. These results are critically examined in Chapter 5, which contextualizes the findings, addresses the inherent limitations of the proposed approach, and highlights avenues for future research. Finally, Chapter 6 summarizes the main contributions of this thesis and provides definitive answers to the central research questions.

2 Background

This chapter establishes the theoretical and geological foundations upon which this research is built. It provides a short overview of the geological setting of the Delft reservoir and a literature review of modern geomodelling and machine learning (ML) techniques. Section 2.1 introduces the geological context of the Delft reservoir, detailing its depositional environment and current conceptual model. Section 2.2 explores the state-of-the-art in geological modelling, comparing traditional techniques and their respective limitations. Finally, Section 2.3 provides a general overview of machine learning and deep learning (DL) principles, focusing on their historical evolution and current application to subsurface characterization. This balanced overview is intended to provide both geological and computer science readers with the necessary context to understand the research methodology and results presented in subsequent chapters.

2.1. The Delft Geothermal Reservoir

The data utilized in this study is sourced from the Delft geothermal reservoir, situated beneath the [Technische Universiteit Delft \(TUD\)](#) Campus in the Netherlands. As of February 2026, a geothermal doublet is actively producing heat from this reservoir (Beerda, 2026). A doublet system consists of a two-well setup: a producer well extracts warm formation water, and an injector well returns the cooled water to the reservoir. At the Delft site, the producer well (**DEL-GT-01**) was drilled to a [True Vertical Depth \(TvD\)](#) of approximately 2,337 m, while the injector well (**DEL-GT-02-S2**) reached a TvD of approximately 2,100 m (Vardon et al., 2024). The project serves both commercial and research objectives, acquiring extensive geophysical logs and core data, supplemented by surface and downhole monitoring instrumentation (Vardon et al., 2020; Vardon et al., 2024). Despite the high data density, the cored section is restricted to the first 80 m of the **DEL-GT-01** well (approximately one-fourth of the reservoir thickness), while **DEL-GT-02-S2** contains no cored sections. The primary target is the [Delft Sandstone \(DSST\)](#), a member of the Lower Cretaceous Nieuwerkerk Formation in the [West Netherlands Basin \(WNB\)](#) (Donselaar et al., 2015).

2.1.1. Geology of the Delft Reservoir

The [WNB](#) is a 60 km wide transtensional basin in the southwest of the Netherlands, characterized by NW-SE trending structural highs and depressions (Donselaar et al., 2015; Ziegler, 1990). The [WNB](#) entered an active rifting stage during the Middle Jurassic (Bodenhausen & Ott, 1981; Den Hartog Jager, 1996; Willems et al., 2020), leading to the formation of several half-grabens filled with terrestrial sediments sourced from neighboring regions (Den Hartog Jager, 1996). The progressive filling of these half-grabens coincided with a change in the depositional environment as a transgressive shoreline moved the system from fluvial to marine. The Nieuwerkerk Formation comprises three members: the Alblasserdam, the [DSST](#), and the Rodenrijs Claystone.

The Alblasserdam member was deposited in a fluvial environment characterized by stacked channels and floodplain deposits (DeVault & Jeremiah, 2002). The [DSST](#) directly overlays the Alblasserdam and consists of thick, stacked fluvial-deltaic channel sandstones with high net-to-gross values, oriented NW-SE (Bruining, 2023; Groenenberg et al., 2010; Mijnlieff, 2020; Willems et al., 2020). The [DSST](#) is in turn overlain by the Rodenrijs Claystone, an organic-rich lagoonal deposit (Bruining, 2023; DeVault & Jeremiah, 2002; Willems et al., 2020).

Based on the sedimentological analysis of the **DEL-GT-01** core and **DEL-GT-02-S2** sidewall samples, three primary facies associations (FAs) have been identified (Bruining, 2023):

- **FA1: Fluvial and distributary channel deposits.** Primarily light grey, fine- to medium-grained sandstones featuring low-angle cross-bedding and ripple lamination, with unit thicknesses ranging from 5 to 80 m.

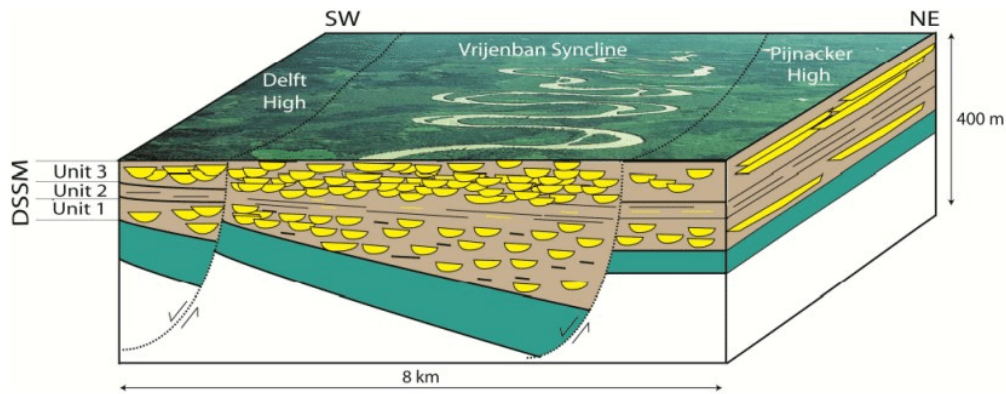


Figure 2.1: Depositional models for the DSST Member with three identified units. Accommodation space is increasing towards the NE (Donselaar et al., 2015)

- **FA2: Crevasse splays and levee deposits.** Composed of moderately sorted, very fine- to medium-grained sandstones interbedded with dark grey siltstones, with thicknesses ranging from 1 to 5 m.
- **FA3: Overbank deposits and paleosols.** Consists of dark grey siltstones, claystones, and very fine-grained sand deposited in low-energy environments. This association is characterized by abundant organic material, slickensides, and siderite nodules.

The specific depositional environment of the DSST remains a subject of scientific discussion. It has been attributed to both fluvial-meandering (Donselaar et al., 2015; Vondrak et al., 2018; Willems et al., 2020) and fluvio-deltaic systems (Bruining, 2023; van Adrichem Boogaert & Kouwe, 1993; Weert et al., 2024). Recent analysis by Bruining (2023) suggests that the cored section represents a deltaic environment. The observed facies associations can be attributed to a upper to lower plain delta with multi-stacked fluvial/distributary channels, single channels, crevasse splays, levees and overbank fines.

2.1.2. Deltaic Depositional Environment

Deltaic environments form at the land-sea interface where river systems supply clastic sediments. Delta morphology is governed by the relative influence of fluvial, wave, and tidal processes (Kolb & Van Lopik, 1966). The subaerial delta plain is typically divided into two domains (Heldreich et al., 2017; Kolb & Van Lopik, 1966):

- **Upper Delta Plain:** Lies above the limit of saltwater intrusion. Sedimentation is driven by distributary channel migration, overbank flooding, and crevasse splaying. Environments include meandering and braided channels, lacustrine fills, and floodplains.
- **Lower Delta Plain:** Subject to river-marine interaction and tidal influence. Channels are more numerous and exhibit bifurcating or anastomosing patterns. Interdistributary bays, bay fills (crevasse splays), and marshes dominate the landscape.

In the DSST, FA1 features organic material and siltstones indicating slackwater periods, while the lack of bioturbation suggests high sedimentation rates. The fine-grained nature of FA2 and FA3 reflects more distal positions on the delta plain. Siderite nodules in FA3 indicate anoxic, waterlogged conditions (Samuel, 1959), while soft-sediment deformation suggests rapid loading of mudstones over saturated sands. These features collectively indicate that FA3 represents deposition within a delta plain, likely in an interdistributary bay or a pro-delta environment.

2.2. Geological Modelling

The primary objective of 3D reservoir modelling is to provide a representative volume of the subsurface for flow simulation and decision support. These models are dynamic, requiring updates as new data becomes available to refine predictions over the reservoir's lifetime (Emerick, 2025; Ringrose & Bentley, 2016). Existing techniques are generally categorized into three types (Koltermann & Gorelick, 1996):

- **Pixel-based Models:** Use spatial statistics (e.g., variograms) for cell-by-cell property assign-

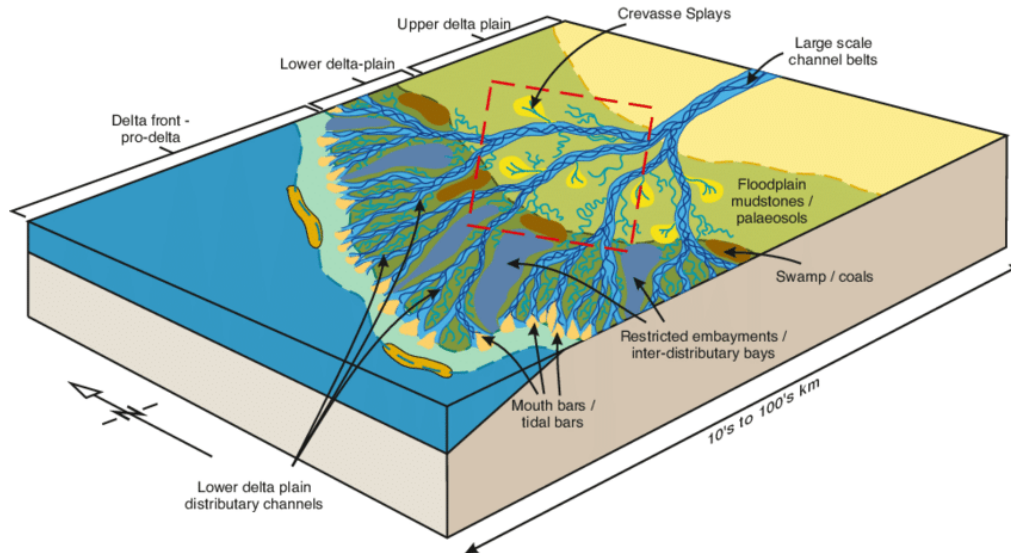


Figure 2.2: Depositional model of the Mungaroo Formation, illustrating the geometries and domains of a fluvial-deltaic system (Heldreich et al., 2017).

ment. They are efficient and honor dense well data but often fail to capture complex connectivity, resulting in pixelated realizations.

- **Object-based Models:** Directly place geometric features (e.g., channels) based on geological rules. They offer high visual realism but struggle to honor dense or closely spaced well data.
- **Process-based Models:** Simulate the processes behind sediment transport and deposition. They provide the highest degree of realism but are computationally expensive and difficult to condition to specific hard data like well logs.

Attribute	Geostatistical	Object-Based	Process-Based
Geological Prior	Variograms, training images.	Geometric shapes and rules.	Physical boundary conditions.
Realism	Low to Moderate.	Moderate to High.	Very High.
Key Advantages	Computational speed, well conditioning.	Realistic connectivity.	Physical consistency.
Key Disadvantages	Lack of geological realistic shapes.	Difficult conditioning.	High computational cost.

Table 2.1: Comparison of spatial modelling techniques in geomodelling.

While traditional pixel-based geomodelling relies heavily on two-point spatial statistics, such as variograms, to distribute properties, these methods often fail to capture the complex, multipoint connectivity inherent in natural geological systems. Machine learning introduces a paradigm shift by moving away from user-defined spatial rules. Instead of requiring an engineer to pre-define the shape of a channel or the range of a variogram, ML algorithms learn these intricate, non-linear spatial relationships directly from data. This makes ML exceptionally well-suited for geomodelling, where the relationships between sparse hard data (e.g., well logs) and the broader geological architecture are too complex to be captured by standard statistical interpolation.

2.3. Machine Learning

Machine learning (ML) involves the development of algorithms that learn patterns from data to perform specific tasks without explicit programming (Goodfellow et al., 2016). This foundational concept

of learning without explicit programming has evolved significantly since Samuel (1959) introduced the term, expanding into a wide array of computational techniques. These diverse techniques are fundamentally categorized based on their training approach into two primary domains:

- **Supervised Learning:** The algorithm is trained on a labeled dataset containing explicit input-output pairs.
- **Unsupervised Learning:** The algorithm identifies hidden patterns within unlabeled data without prior knowledge of the desired output.

Within these two broad categories, algorithms are deployed to solve common tasks such as classification, regression, clustering, and generation. Applying these generative tasks to the subsurface highlights a severe limitation in traditional pixel-based geomodelling, which relies heavily on simple two-point spatial statistics (e.g., variograms) to distribute properties. Because these simple statistics often fail to capture the complex, multipoint connectivity inherent in natural geological systems, ML introduces a necessary paradigm shift. This shift moves the modelling workflow away from strict, user-defined spatial rules. Instead of requiring an engineer to pre-define the range of a variogram, ML algorithms learn intricate, non-linear spatial relationships directly from the data. Learning directly from data makes ML exceptionally well-suited for geomodelling, where the relationships between sparse hard data (e.g., well logs) and the broader geological architecture are simply too complex to be captured by standard statistical interpolation. The capacity to uncover such complex, non-linear relationships in vast datasets is largely driven by modern [Artificial Neural Networks \(ANNs\)](#) (LeCun et al., 2015). Inspired by the biological structure of the brain, these [ANNs](#) consist of interconnected artificial neurons that process information across multiple computational layers (Goodfellow et al., 2016).

While [ANNs](#) and traditional machine learning algorithms excel at analyzing tabular petrophysical data for various subsurface tasks, they inherently lack actual geological reasoning skills. Without such reasoning to naturally make sense of areas with low data quality, these algorithms are restricted to inferring abstract features directly from their training data ([schaaf2019quantification](#)). Inferring these features in standard ML workflows necessitates manual feature extraction, a process where engineers must explicitly define the relevant geological parameters prior to training. This strict reliance on manual feature extraction creates a critical limitation when attempting to scale these algorithms to 3D geomodels. Within these high-resolution 3D geomodels, which typically contain millions of voxels, manually defining multi-scale spatial features ultimately becomes computationally impractical.

Deep Learning (DL) overcomes this limitation through automatic feature extraction. By utilizing multi-layered networks, DL architectures natively process high-dimensional spatial grids, bypassing the manual extraction bottleneck. For subsurface characterization, this means a DL model can ingest raw 3D conceptual models or process-based simulations and autonomously identify the underlying geological rules. This capability to scale with high-dimensional data makes DL the premier mathematical framework for modern reservoir modelling.

2.3.1. Deep Learning and Inductive Biases

Deep learning (DL) is a subset of ML focusing on multi-layered neural networks that represent data with multiple levels of abstraction (Goodfellow et al., 2016; LeCun et al., 2015). Achieving these levels of abstraction in high-dimensional spaces introduces a fundamental challenge known as the “curse of dimensionality.” This curse of dimensionality describes the exponential increase in data required as the spatial dimensions grow, making high-dimensional estimation computationally challenging. To overcome this computational challenge and help models generalize from limited data, engineers implement specific architectural choices known as inductive biases (Prince, 2023). A premier example of such an inductive bias is the foundational architecture used for processing spatially structured data: the [Convolutional Neural Network \(CNN\)](#).

[CNNs](#) leverage spatial inductive bias to become highly effective for 2D and 3D geomodelling tasks. Executing these spatial tasks relies on a core architectural innovation known as the convolutional layer. This convolutional layer replaces the fully connected neurons of a standard [ANN](#) with a localized set of learnable filters, commonly referred to as kernels. These multi-dimensional kernels operate by systematically sliding across the input data in discrete increments dictated by a predefined step size, or stride. This sliding stride is frequently combined with padding, a technique that adds artificial zero-

value borders to the input to ensure the entire spatial region is comprehensively analyzed. Analyzing this spatial region involves computing a local dot product between the kernel weights and the input values, a mathematical operation visually illustrated in Figure 2.3.

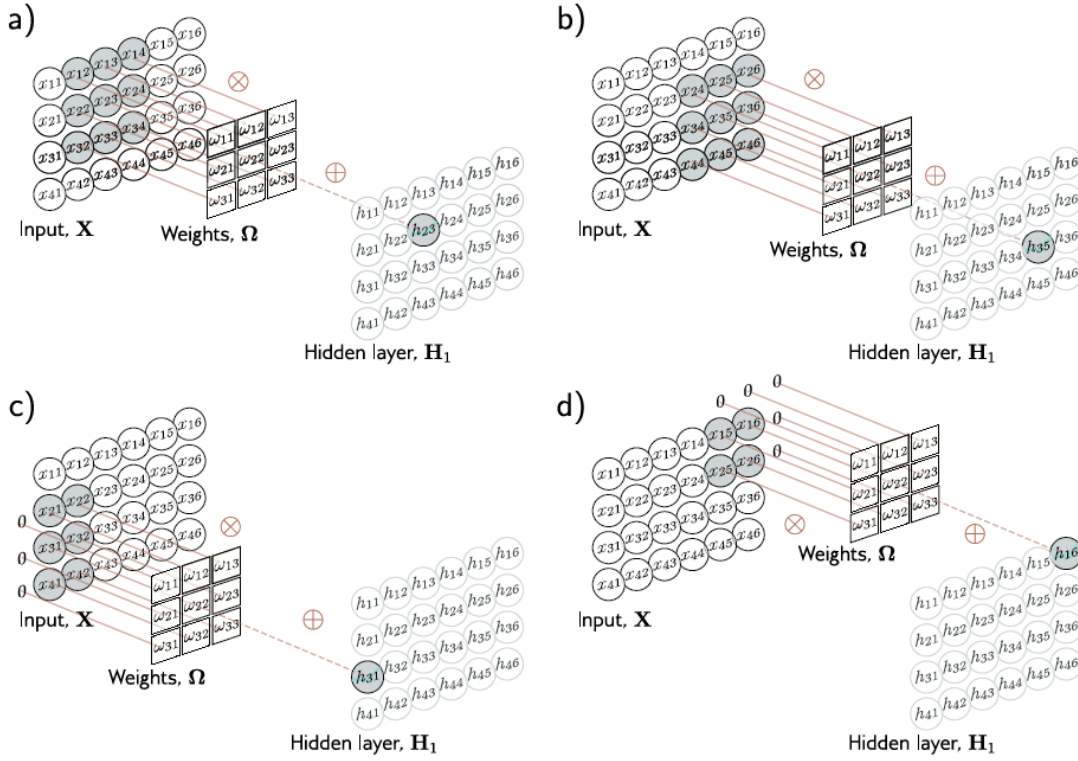


Figure 2.3: Example of a 2D convolutional layer with kernel size of 3x3 and a padding of 1. Each output h_{ij} is composed of a weighted sum of the 3x3 nearest inputs on the previous layer (Prince, 2023). a) Here, the output h_{23} (shaded output) is a weighted sum of the nine positions from x_{12} to x_{34} (shaded inputs). b) Different outputs are computed by translating the kernel across the image grid in two dimensions. c–d) With zero-padding, positions beyond the image's edge are considered to be zero.

The mathematical result of this operation is a new, filtered spatial representation known as a feature map. This feature map is typically downsampled by a subsequent pooling layer, which condenses the spatial dimensions while preserving the most prominent information. Preserving this information across a continuous stack of convolutional and pooling layers allows the network to progressively map low-dimensional input data into high-dimensional, abstract features. At the base of this feature mapping process, the initial layers capture simple, low-level spatial patterns, such as discrete edges or sharp facies boundaries. By mathematically combining these simple low-level edges in deeper layers, the network learns to recognize highly complex, high-level geological architectures, such as the overall sinuosity of a fluvial channel. Learning these complex high-level architectures is computationally efficient because the exact same kernel is applied universally across the entire grid, creating an inherent advantage known as translation invariance. This translation invariance ensures that if the network learns to identify a channel structure in one corner of the reservoir, it can accurately recognize that same structure anywhere else in the model, drastically reducing computational overhead and preventing overfitting. Further preventing this overfitting requires another critical component of the DL workflow: the strict use of validation and testing protocols. These validation protocols ensure that the hierarchical patterns learned by the network are robust and safely applicable to unseen data. Applying these robust, learned spatial patterns to unseen subsurface data forms the fundamental basis of modern **Deep Learning (DL)**-driven geomodelling workflows.

2.3.2. Geomodelling with Deep Learning

Modern DL-driven geomodelling workflows have evolved significantly, transitioning from simple pixel-based interpolations to the generation of highly sophisticated spatial representations. Generating these sophisticated representations relies on the ability of DL to directly learn complex geological features from large datasets or process-based simulations. Learning directly from such simulations bridges the historical gap between simplistic descriptive statistics and true geological realism (Laloy et al., 2017). To achieve this realism, CNNs have proven particularly successful in capturing the complex, non-stationary architectures of fluvial channels and facies distributions (Sun et al., 2023). Capturing these intricate architectures enables the rapid generation of stochastic realizations that successfully honor both the global geological concept and local, sparse conditioning data. Honoring these dual constraints while generating massive ensembles of stochastic realizations fundamentally enhances subsurface uncertainty characterization (Song et al., 2023). This unique capacity for rapid, constrained generation is primarily driven by specialized neural network frameworks, most notably generative architectures.

These generative architectures that are using CNNs, are best known for their ability to generate realistic images. The most-well known models include VAEs, GANs and Latent Diffusion Models. Though they are all different in terms of architecture, some core concepts remain similar. Generative models are designed to learn and approximate the complex joint probability distribution of the training data whereby elements are generated based on previously generated values (He2025generative). Moreover, these models use a latent space, designed to either store the encoded training data (VAEs) or be used to scale to an image through various upsampling steps (GANs and Latent Diffusion Models) (Prince, 2023).

2.3.3. Generative Adversarial Networks

Generative architectures, specifically Generative Adversarial Networks (GANs), achieve complex spatial generation through a competitive framework between two distinct neural networks (Goodfellow et al., 2014). These two networks are defined as a Generator (**G**) and a Discriminator (**D**). The generator is tasked with creating synthetic spatial samples from random noise, while the discriminator is simultaneously tasked with distinguishing these synthetic samples from actual real-world training data. The continuous competition to outsmart one another drives a minimax game, which is mathematically expressed as:

$$\min_{\mathbf{G}} \max_{\mathbf{D}} V(\mathbf{D}, \mathbf{G}) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log \mathbf{D}(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - \mathbf{D}(\mathbf{G}(\mathbf{z})))] \quad (2.1)$$

In this equation, $\mathbf{x}$ represents the real data, and $\mathbf{z}$ represents a random latent vector sampled from a prior distribution $p_{\mathbf{z}}$. The first term represents the discriminator successfully recognizing real geological data, while the second term represents the discriminator successfully catching fake data generated by $\mathbf{G}(\mathbf{z})$. During training, the discriminator seeks to maximize this overall value function to perfect its detection abilities, while the generator seeks to minimize it to successfully fool the discriminator. By successfully fooling the discriminator, the generator ultimately learns to produce high-fidelity geological realizations without requiring any explicit, hard-coded spatial rules.

While generating high-fidelity, unconstrained realizations is a powerful capability, practical reservoir modelling requires these models to honor sparse physical constraints, such as hard well data. Honoring this hard data is typically achieved through an optimization technique known as GAN inversion. Instead of altering the neural network architecture itself to accommodate well logs, GAN inversion operates strictly within the model's latent space after the training phase is complete. By iteratively searching and optimizing this latent space, the algorithm identifies a specific latent vector $\mathbf{z}$ that generates a 3D volume perfectly matching the sparse well data (Rongier & Peeters, 2025b). Identifying this optimal vector allows engineers to securely condition the generative model to real-world subsurface observations.

Once these conditioned realizations are generated, their geological realism and consistency must be rigorously evaluated. Evaluating and validating high-dimensional 3D spatial data is historically computationally difficult due to the curse of dimensionality. To overcome this computational difficulty, recent geomodelling workflows utilize refined validation metrics such as the Mean Sliced Wasserstein Distance (MSWD) (Bhavsar et al., 2024). Rather than comparing massive 3D grid volumes directly, MSWD bypasses the high dimensionality by mathematically projecting the 3D facies distributions onto

multiple 1D slices. Comparing the distances between these 1D slices provides a highly efficient and robust metric to validate whether the newly generated geomodels remain geologically consistent with the original training data.

3 Methodology

This chapter describes the comprehensive project workflow, beginning with the generation of 3D synthetic datasets used for training, validating, and testing the models (Section 3.1). Section 3.2 details the GAN framework, including the network architectures, the facies representation strategy, and the loss functions employed during training. Finally, Section 3.2.2 introduces the metrics and simulation methods used to evaluate the geological realism and practical utility of the generated realizations.

3.1. Dataset Generation

In order to rigorously test the applicability of GANs several new datasets had to be generated. Training deep learning algorithms for 3D geomodelling requires a substantial volume of high-fidelity training data. High-fidelity training data is inherently scarce due to the high uncertainty and limited availability of subsurface measurements, making synthetic data generation via stochastic process-based algorithms the established approach to provide sufficient training samples (Rongier & Peeters, 2025b; Sun et al., 2023). These new datasets were generated using FLUMY (Martin et al., 2016), a process-based stochastic modelling software designed to reproduce meandering channelized reservoirs. Meandering channelized reservoirs are accurately replicated because FLUMY simulates facies, relative grain size, and depositional time for each voxel in a 3D grid based on specified hydraulic and sedimentological rules. These specified hydraulic and sedimentological rules ensure that the models the software generates contain geologically realistic features. Geologically realistic features are produced without requiring excessive computational overhead, making the software not very computationally intensive and making it possible to generate a large number of models in a relatively short amount of time. This rapid generation capacity is one of the primary characteristics why FLUMY has already successfully been implemented in the research by Bhavsar et al. (2024) to generate samples for training GANs.

Training samples for these generative frameworks can be tailored because FLUMY is designed to be compatible with a wide range of input parameters, of which a full list can be found on their [website](#). This full website list is simplified in the python distribution, which contains so-called Non-Expert User Settings (NEXUS) that only allow three customizable parameters: 1) Maximum Channel Depth, 2) Maximum Sand Body Extension, and 3) Net to Gross. Based on these three customizable inputs, the full list of parameters as provided in the [user guide](#) is estimated automatically. Although the software is specifically designed to simulate fluvial meandering rivers, its underlying process-based parameters can be intentionally adapted to mimic alternative depositional domains. Mimicking these alternative depositional domains allows the software to effectively capture the complex deltaic architectures hypothesized for the Delft Sandstone reservoir, establishing a strong foundation for the specific configurations of the underlying conceptual models.

3.1.1. Dataset Descriptions

The three distinct geological conceptual models developed to capture the range of environments hypothesized for the Delft Sandstone reservoir are compiled into separate training suites. These separate training suites consist of Dataset A, Dataset B, and Dataset C. Dataset A (Upper Plain Deltaic) initiates this sequence and is characterized by meandering fluvial channels and extensive floodplain deposits. Extensive floodplain deposits and deep meandering channels stand in direct contrast to the configurations within Dataset B (Lower Plain Deltaic). Dataset B (Lower Plain Deltaic) reflects an increased marine influence with shorter, more numerous distributary channels. Shorter, more numerous distributary channels are subsequently blended with upper plain characteristics to construct Dataset C (General Deltaic). Dataset C (General Deltaic) utilizes a stochastic combination of both settings to represent a transitional depositional system.

The hypothesized depositional environments are split across three discrete suites, starting with Dataset A (Upper Plain Deltaic), which is characterized by meandering fluvial channels and extensive floodplain

deposits. Extensive floodplain deposits and deep meandering channels stand in contrast to Dataset B (Lower Plain Deltaic), a setting that reflects an increased marine influence with shorter, more numerous distributary channels. Shorter, more numerous distributary channels are subsequently blended with upper plain characteristics to construct Dataset C (General Deltaic), which utilizes a stochastic combination of both settings to represent a transitional depositional system.

This transitional depositional system and the overall geological variability across all three datasets are explicitly controlled by three primary NEXUS input parameters: Net-to-Gross (NTG), maximum channel depth, and maximum sand body extension. Maximum sand body extension, maximum channel depth, and target NTG values define the explicit architectural bounds of each setting, as summarized in Table 3.1.

Parameter	Dataset A	Dataset B	Dataset C
Net-to-Gross (Max. Channel Depth (m))	6	4	4–6
Max. Sand Body Extension	100	60	60–100

Table 3.1: NEXUS input parameters for the three training datasets.

Beyond these specific NEXUS constraints, all generated models share a uniform reservoir scale with a total grid resolution fixed at $128 \times 128 \times 32$ voxels. Establishing this specific grid resolution combined with a standard voxel dimension of $20 \text{ m} \times 20 \text{ m} \times 1 \text{ m}$ means that the total simulated domain securely covers a volume of $2560 \text{ m} \times 2560 \text{ m} \times 32 \text{ m}$. A simulated volume with a 32 m grid depth functions as a highly focused reservoir interval where the sparse conditioning challenge is most acute, even though individual channel sandstone units within the broader regional stratigraphy can vertically stack up to an extreme thickness of 80 m. Every single voxel within this acute 32 m vertical interval is ultimately encoded using a specific categorical index chosen from a suite of 9 primary facies classes.

Value	Facies Key	Color	Description
0	Background		Undefined/Muddy matrix
1	Channel Lag		Coarse active channel fill
2	Point Bar		Meandering lateral accretion
3	Sand Plug		Fine-grained channel abandonment
4	Crevasse Splay		Proximal high-energy overbank
5	Crevasse Channel		Feeder for splay deposits
6	Levee		Bordering channel ridges
7	Overbank		Vegetated floodplain silt
8	Wetland/Paleosol		Organic-rich anoxic sediments

Table 3.2: Overview of the 9 facies classes used in the training datasets.

3.2. Fluvgan Framework

The **Fluvial Generative Adversarial Network (FluvGan)** model, introduced by Rongier and Peeters (2025a), is a conditional GAN architecture optimized for 3D fluvial facies modelling. It builds upon the BigGAN framework (Brock et al., 2019) to handle the high resolution and complex connectivity required for reservoir models. Previous studies have demonstrated that **FluvGan** can generate geologically plausible realizations while remaining training-stable (Rongier & Peeters, 2025b). Crucially, Rongier and Peeters (2025c) showed that the model’s latent space can be inverted and optimized to honor hard constraints from well data.

3.2.1. Model Architecture

The architecture comprises a Generator (**G**) and a Discriminator (**D**) using 3D Residual blocks. To handle the categorical nature of facies data, the project employs a one-hot encoding strategy where

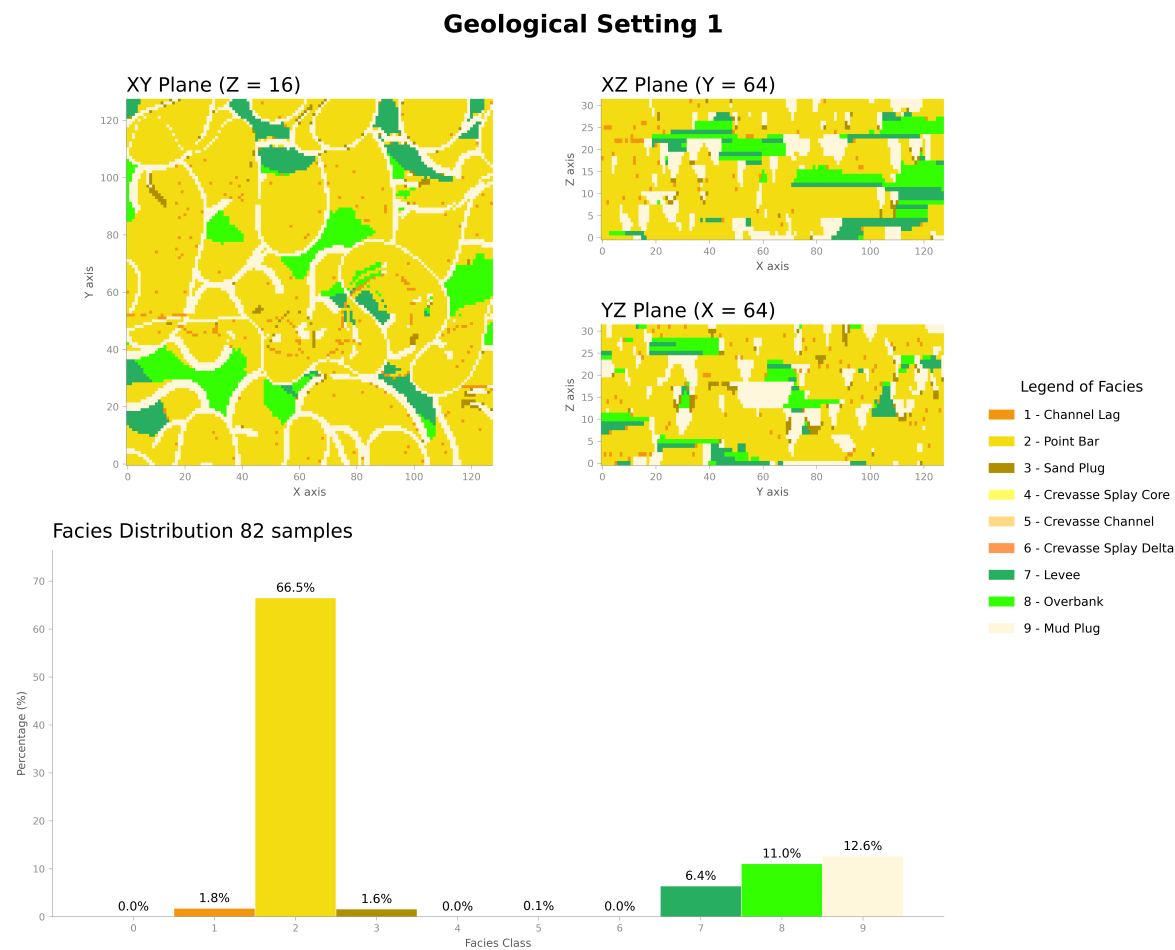


Figure 3.1: Visualization of training data from Dataset A. The rows represent facies distribution, relative grain size, and depositional time, respectively.

each facies is represented as a separate channel in the output volume. The generator transforms a 100-dimensional latent vector $\mathbf{z}$ into a $9 \times 32 \times 128 \times 128$ volume. It utilizes a 3D transposed convolution to project the latent vector into a $1024 \times 4 \times 4 \times 4$ feature map, followed by five residual upscaling blocks. To accommodate the anisotropy of the grid, spatial dimensions are doubled in each block (up to 128), while the depth dimension is doubled only in the first three blocks (up to 32). The final layer uses a Softmax activation to ensure the 9 facies channels sum to one per voxel. The discriminator acts as a binary classifier, mapping the $9 \times 32 \times 128 \times 128$ input back to a single scalar logit. It consists of five residual downscaling blocks that iteratively reduce the spatial resolution while increasing the channel count to 1024. Both networks employ spectral normalization and LeakyReLU activations to ensure stable training. The generator and discriminator architectures contain approximately 11.6 and 2.0 million trainable parameters, respectively.

Hyperparameter	Value	Description
Latent Dimension ($\dim(\mathbf{z})$)	100	Gaussian prior
Batch Size	16	Samples per iteration
Learning Rate (G, D)	$10^{-4}, 4 \cdot 10^{-4}$	Two-timescale update rule
Kernel Size	$3 \times 3 \times 3$	3D Convolutional filters
Optimizer	Adam	$\beta_1 = 0, \beta_2 = 0.999$

Table 3.3: Key hyperparameters for the Fluvgan architecture.

3.2.2. Conditioning and Evaluation

To honor sparse well data, a latent space optimization approach is used. The latent vector $\mathbf{z}$ is iteratively refined to minimize the difference between the generated facies at well locations and the true observations. The realism of the final realizations is evaluated using the Mean Sliced Wasserstein Distance (MSWD) across 1,000 projections, ensuring that the multi-point statistics and connectivity of the training data are preserved (Bhavsar et al., 2024).

Shannon’s entropy fomrula:

$$H(X) = - \sum_{i=1}^n P(x_i) \log_2 P(x_i) \quad (3.1)$$

4 Results

4.1. Post Training Results

4.2. Post Conditioning Results

5 Discussion & Recommendations

5.1. Interpretations of key results

5.2. Limitations of the current work

5.3. Recommendations for future work

6 Conclusion

6.1. Answers to Research Questions

6.2. Final Remarks

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